

Karman Sidhu

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EDUCATION

McMaster University – Bachelor of Engineering (Co-op)
Mechatronics Engineering

Hamilton, Ontario
Sept. 2024 - May 2029

Relevant Coursework: Embedded Systems, Analog and Digital Circuits, Mechanical Engineering Measurements and Design, Mechanics (Dynamics & Statics), Signals & Systems, Software Development, Data Structures & Algorithms

SKILLS

Embedded & Electrical Systems: Arduino, Microcontrollers, Sensor Integration, PID Control, PWM, ADC, Digital Logic, SPI/I2C/UART, Oscilloscope, Finite State Machines (FSM), Hardware Testing

Programming: Python, C++, C, Java, HTML/CSS, JavaScript, SQL, Object-Oriented Programming (OOP)

Software & Tools: SolidWorks, MATLAB, Git/GitHub, Node.js, Express.js, MongoDB, REST APIs, Multisim, Excel

Manufacturing: CNC Machining, Carbon Fiber Composites, Soldering, Power Tools, 3D Printing, Laser Cutting

PROJECTS

Embedded Thermal Regulation & Cooling Control System

March 2026 – May 2026

- Designed closed-loop system using Arduino Uno (C++), LM35 temperature sensor, LM358 signal conditioning, ADC feedback, PWM-controlled fan, MOSFET driver, optocoupler isolation, and 4-state FSM logic for real-time temperature regulation.
- Integrated 8+ hardware/software components through embedded programming, calibration, system testing, debugging, and circuit design; implemented a 100 ms (10 Hz) control loop, 15-30°C adjustable temperature threshold, RC noise filtering, and PWM speed ramping for stable operation.

Maintry – Personal Asset Maintenance Platform

Feb. 2026 – July 2026

- Developed a full-stack web application using Node.js, Express.js, MongoDB, and JavaScript featuring 8+ REST API endpoints for asset management, scheduling, and maintenance tracking.
- Developed secure backend using CRUD operations, JWT authentication, bcrypt, and MongoDB integration.
- Integrated Google Places API to retrieve and filter service professionals, ranking the top 10 results based on ratings, location, and contact information.

Airport Baggage Sortation Mechanism

Oct. 2024 – Jan. 2025

- Designed and iterated a baggage-transfer mechanism in Autodesk Inventor using hinged ramps, rack-and-pinion motion, and custom 3D-printed components, optimizing mechanical assembly and functionality.
- Integrated a Raspberry Pi-controlled rotary actuator and Q-Arm; created engineering assemblies and drawings, refining geometry through iterative prototyping and testing.

EXPERIENCE

Data Clerk (Contract, Full-Time) – TJX Canada

May 2025 – Aug. 2025

- Processed and validated operational records daily from multiple departments, ensuring accurate data entry and quality control.
- Analyzed data inconsistencies, verified records, and collaborated with internal teams to resolve issues and maintain data integrity and workflow efficiency.
- Utilized internal databases for data verification, reporting, and record management.

LEADERSHIP & INVOLVEMENT

MAC Engineers Without Borders – VP of Media & Former Conference Co-Director

Sept. 2024 – Present

- Co-led MAC-IMPACT conference planning, logistics, operations, and speaker outreach across multiple universities, engaging 75+ students and professionals.
- Led technical workshops on robotics and water treatment with EWB Toronto Professional Chapter.
- Manage a 5-person media team to develop communication and branding strategies, launch the team's first website, and generate 10,000+ average views and engagements across digital platforms.

McMaster Sumobot Club – Web Development Executive

Aug. 2026 – Present

- Build registration and team portal using HTML/CSS/JS; process participant data via Python and Excel
- Maintain main team website and develop competition features to streamline events and workshop sign-ups.

MAC Formula Electric SAE – Manufacturing Team Member

Sept. 2024 – Sept. 2025

- Supported chassis and composite monocoque fabrication through carbon fiber layups, foam core shaping, vacuum resin infusion, structural assembly, and quality-controlled manufacturing processes for a racing vehicle.
- Machined precision parts (e.g. mounts) and supported accumulator cart and aerodynamic manufacturing.